

K100 ROS 2 Motion Planning

本项目包含 K100 机器人的运动规划功能，支持 BrainCo 和 RoHand 两种灵巧手配置。

1. 编译与环境配置

编译工作空间

```
colcon build --packages-select k100_motion_planning
# 或者编译所有包
colcon build
```

配置环境变量

```
# 如果使用 zsh
source install/setup.zsh

# 如果使用 bash
source install/setup.bash
```

2. 启动节点

步骤 1: 启动 MoveIt 2 和 RViz

根据您的手型配置选择启动命令：

BrainCo 灵巧手 (默认):

```
ros2 launch k100_brainco_moveit_config demo.launch.py
# 若无需 RViz 可视化界面:
ros2 launch k100_brainco_moveit_config demo.launch.py use_rviz:=false
```

RoHand 灵巧手:

```
ros2 launch k100_rohand_moveit_config demo.launch.py
# 若无需 RViz 可视化界面:
ros2 launch k100_rohand_moveit_config demo.launch.py use_rviz:=false
```

步骤 2: 启动运动规划节点

```
# 默认 (BrainCo 手型)
ros2 launch k100_motion_planning k100_motion_planning.launch.py

# 指定 RoHand 手型
ros2 launch k100_motion_planning k100_motion_planning.launch.py
hand_name:=k100_rohand
```

注意: 规划节点启动时, 关于 `right_hand`、`left_hand` 以及 `other_joints` 的报错可以忽略。

```
[x100_motion_planning_node-1] [INFO] [1757929419.098790921] [x100_motion_planning_node.moveit.core.robot_model]: No root/virtual joint specified in SRDF. Assuming fixed joint
[x100_motion_planning_node-1] [INFO] [1757929419.575318582] [x100_motion_planning_node.moveit.kinematics.kdl_kinematics_plugin]: Joint weights for group 'left_arm': 1 1 1 1 1 1
[x100_motion_planning_node-1] [INFO] [1757929419.575410227] [x100_motion_planning_node.moveit.kinematics.kdl_kinematics_plugin]: Joint weights for group 'right_arm': 1 1 1 1 1 1
[x100_motion_planning_node-1] [ERROR] [1757929419.575425707] [x100_motion_planning_node.moveit.kinematics.kdl_kinematics_plugin]: Group 'left_hand' is not a chain
[x100_motion_planning_node-1] [ERROR] [1757929419.575443952] [x100_motion_planning_node.moveit.ros.kinematics_plugin_loader]: Kinematics solver of type 'kdl_kinematics_plugin/KDLKinematicsPlugin' could not be initialized for group 'left_hand'
[x100_motion_planning_node-1] [ERROR] [1757929419.575450554] [x100_motion_planning_node.moveit.ros.robot_model_loader]: Kinematics solver could not be instantiated for joint group left_hand.
[x100_motion_planning_node-1] [ERROR] [1757929419.575461605] [x100_motion_planning_node.moveit.kinematics.kdl_kinematics_plugin]: Group 'right_hand' is not a chain
[x100_motion_planning_node-1] [ERROR] [1757929419.575476764] [x100_motion_planning_node.moveit.ros.kinematics_plugin_loader]: Kinematics solver of type 'kdl_kinematics_plugin/KDLKinematicsPlugin' could not be initialized for group 'right_hand'
[x100_motion_planning_node-1] [ERROR] [1757929419.575479139] [x100_motion_planning_node.moveit.ros.robot_model_loader]: Kinematics solver could not be instantiated for joint group right_hand.
[x100_motion_planning_node-1] [ERROR] [1757929419.575488857] [x100_motion_planning_node.moveit.kinematics.kdl_kinematics_plugin]: Group 'other_joints' is not a chain
[x100_motion_planning_node-1] [ERROR] [1757929419.575491002] [x100_motion_planning_node.moveit.ros.kinematics_plugin_loader]: Kinematics solver of type 'kdl_kinematics_plugin/KDLKinematicsPlugin' could not be initialized for group 'other_joints'
[x100_motion_planning_node-1] [ERROR] [1757929419.575492885] [x100_motion_planning_node.moveit.ros.robot_model_loader]: Kinematics solver could not be instantiated for joint group other_joints.
[x100_motion_planning_node-1] [INFO] [1757929419.595259623] [x100_motion_planning_node.moveit.ros.move_group]
```

3. 服务调用测试

3.1 关节空间规划 (PlanJointGoal)

单臂 (Right Arm):

```
ros2 service call /plan_joint_goal k100_motion_planning/srv/PlanJointGoal "
{group_name: 'right_arm', joint_angles: [0.0, -0.5, 0.0, -1.0, 0.0, 0.0,
0.0]}"
```

双臂 (Both Arms):

注意: 关节数组顺序通常为 [左臂关节..., 右臂关节...], 共 14 个关节。

```
ros2 service call /plan_joint_goal k100_motion_planning/srv/PlanJointGoal "
{group_name: 'both_arms', joint_angles: [0.0, 0.7, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]}"
```

3.2 播放 Rosbag 轨迹 (PlayRosbag)

支持时间参数化优化 (TimeOptimalTrajectoryGeneration)。

```
ros2 service call /play_rosbag k100_motion_planning/srv/PlayRosbag "{group_name: 'left_arm', file_path: '/home/x100/wh/k100_ros2/src/k100_motion_planning/bag/rosbag2_2025_12_03-15_49_22/rosbag2_2025_12_03-15_49_22_0.db3', enable_time_parameterization: true}"
```

3.3 笛卡尔空间规划 (PlanPoseGoal)

```
ros2 service call /plan_pose_goal k100_motion_planning/srv/PlanPoseGoal "{group_name: 'left_arm', pose: {header: {frame_id: 'base_link'}, pose: {position: {x: 0.22356, y: 0.21, z: 0.87223}, orientation: {x: 0.6145, y: -0.61441, z: 0.34989, w: -0.3498}}}}"
```

3.4 逆运动学求解 (ComputeIK)

```
ros2 service call /compute_ik k100_motion_planning/srv/ComputeIK "{group_name: 'left_arm', pose: {header: {frame_id: 'base_link'}, pose: {position: {x: 0.22356, y: 0.21, z: 0.87223}, orientation: {x: 0.6145, y: -0.61441, z: 0.34989, w: -0.3498}}}}"
```